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MDR_2439-EXNA_RATPLASMA	EXECUTION OF THE PNA-HPLC-ASSAY FOR DETERMINATION OF 2439-EXNA FROM RAT PLASMA (METHOD DEVELOPMENT REPORT)	
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1 RELATED DOCUMENTS

This instruction describes the execution of a PNA-HPLC-Assay based on the following documents:

- KU-RGL-SOP-002 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-002 (formerly: REG-102-S02) – Method Validation Data Report
  - KU-RGL-ESS-005 (formerly: REG-102-S05) – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot
- KU-RGL-SOP-009 (formerly: REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-013 (formely: REG-109-S03) – PNA Assay Instruction and Protocol – Liquid Matrices - OnePot
  - KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions
  - KU-RGL-INS-005 (formerly: REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGL-SOP-010 (formerly: REG-110) – Herstellung von Standardlösungen mit einer Präzisionswaage (*Preparation of Standard Solutions Using a Precision Balance*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
- Development protocol in the project folder AN400.0-24 under the following link:

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The above mentioned documents describe the assay execution under nonGLP-relevant conditions. The parameters and results described in this document are generated from a nonGLP method development, which was conducted following the above mentioned SOPs.

2 DEFINITIONS AND ABBREVIATIONS

°C	Grad Celsius
AB0X	Analytical batch AN400.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
ed	Edition
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HPLC	High Performance Liquid Chromatography
ICH	International Conference on Harmonization
IFU	Instructions For Use
K2-EDTA	Dipotassium Ethylenediaminetetraacetic Acid
LC/MS	Liquid Chromatography / Mass Spectrometry
LLOQ	Lower Limit of Quantitation
min	Minute
n.a.; N/A; N.A.	Not applicable
PBS	Phosphate Buffered Saline
PI	Principal Investigator
PNA	Peptide Nucleic Acid
QC	Quality Control
rcf	Relative centrifugal force
RNA	Ribonucleic Acid
siRNA	Small Interfering Ribonucleid Acid
SOP	Standard Operating Procedure
SD	Standard Deviation
SST	System Suitability Test(ing)
ULOQ	Upper Limit of Quantitation

3 ACTIVITIES TO BE PERFORMED

This method development report describes an anion exchange (AEX)-HPLC method with fluorescence detection that allows the specific detection of the analyte 2439-exNA from rat plasma. 2439-exNA is a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with the sequence depicted in Figure 1:

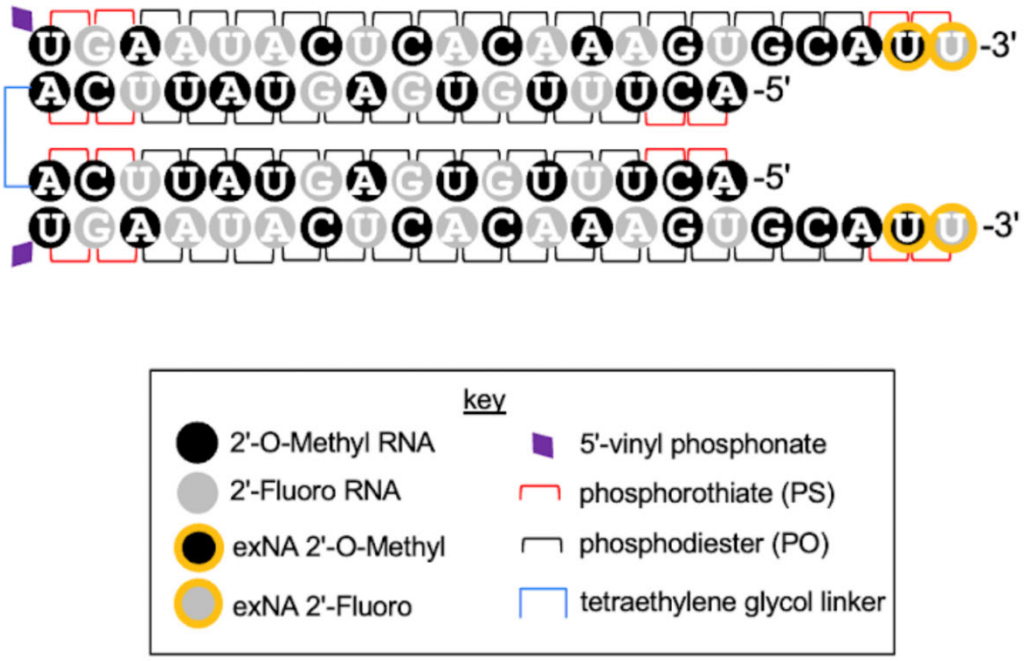

Figure 1 Structure and sequence of the analyte 2439-exNA

The assay is based on the specific hybridization of the antisense strands of 2439-exNA to the complementary PNA-probe. The PNA-probe is N-terminally conjugated with an Atto425-fluorescent dye. The utilized PNA-BroadInst-01 is complementary to the antisense strand of 2439-exNA and has the following sequence:

N-term Atto425-OO-cdc ttt gtg dgt d C-term

This method was developed without having any metabolites of 2439-exNA.

Figure 2 shows the HPLC-profile of 2439-exNA in rat plasma.

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4 RESULTS FROM QUALIFICATION RUN IN RAT PLASMA

In this section, the raw data for the determination of calibration curve accuracy (Table 2), confirmation of LLOQ (Table 3), selectivity (Table 4), carry-over (Table 5), accuracy and precision (Table 6), freeze and thaw and short term stability (Table 7) and dilution integrity (Table 8) from 2439-exNA in rat plasma are shown. The results were generated in one qualifying run. Calibration standard and QC samples were prepared by spiking 2439-exNA into rat plasma. Samples were prepared according to the assay procedure described in Section 5.

During the qualification run, the calibration range was between 1.0 ng/mL and 1,000.0 ng/mL. All calibration standards were within ±15 % (±20 % at the LLOQ) of the nominal concentration.

Selectivity and Sensitivity

The LLOQ was confirmed at 1.0 ng/mL in rat plasma. No signal was detected in corresponding blank plasma samples and the 2439-exNA peaks at the LLOQ of 1.0 ng/mL were at least 5 times the background signal in the analysis run. At the LLOQ, the mean back-calculated concentration was 105.0 % of nominal with a CV of 6.7 %.

The selectivity of the method was demonstrated by comparison of unspiked samples of rat plasma with the same plasma samples spiked with 2439-exNA at a concentration of the LLOQ. The signal for 2439-exNA in the spiked samples gave an at least 5-fold higher response (peak area) compared to the blank samples.

Carry Over

Carry over plasma samples were run after the 1,000.0 ng/mL calibration standards. There was no distinct analyte signal detectable in the carry-over samples after the 1,000.0 ng/mL standards.

Accuracy and Precision

QC samples were prepared at 4 concentration levels which are covering the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- at 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at around 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

The QC samples for the determination of the accuracy and precision of the method met the acceptance criteria with following results:

- 91.7 % (11 out of 12) of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies at the 4 concentration levels were 106.7 % (QC-LLOQ), 86.7 % (QC-Low), 102.3 % (QC-Mid) and 103.5 % (QC-High).
- The mean precisions at the 4 concentration levels were 5.4 % (QC-LLOQ), 3.8 % (QC-Low), 3.3 % (QC-Mid) and 0.2 % (QC-High).

Freeze-Thaw and Short-Term-Stability

The freeze and thaw (maximum of three cycles) and short term stability (once for 4 hours on wet ice) of 2439-exNA in rat plasma was successfully tested. Working solutions for stability samples were prepared in glass vials at room temperature and spiked and aliquoted on ice. Aliquots of the 2439-exNA at the 2 QC concentration levels QC-Low and QC-High were frozen at -80°C. After thawing the samples were stored on wet ice for 4 hours and frozen again. The samples were analyzed in triplicate.

After 3 freeze/thaw cycles and 4 hours on wet ice the mean back-calculated concentrations were at 95.6 % (QC-Low) and 106.5 % (QC-High) of the nominal value.

Dilution Integrity

The precision and accuracy of rat plasma sample dilution was evaluated using a DI-QC standard prepared at concentration levels 1,000-fold (500,000 ng/mL) above the ULOQ of 1,000.0 ng/mL. This QC sample was analyzed and quantified after 1:1,000 dilution in 3 replicates. (see below and section 5.4.3.1). The mean accuracy was 105.3 % and the CV was 1.2 %.

During method development different buffer solutions were tested.

Table 2 Calibration Curve Accuracy in K2 EDTA rat plasma

2439-exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	Sample No.	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-20 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal
AB07	13 - 18	1.1	110.0%	4.7	94.0%	50.0	100.0%	101.5	101.5%	503.6	100.7%	992.8	99.3%
AB07	49 - 54	1.0	100.0%	4.7	94.0%	51.5	103.0%	99.0	99.0%	494.7	98.9%	1,007.4	100.7%
Mean		1.05	105.0%	4.70	94.0%	50.75	101.5%	100.25	100.3%	499.15	99.8%	1,000.10	100.0%
SD (+/-)		0.07		0.00		1.06		1.77		6.29		10.32	
CV (%)		6.7%		0.0%		2.1%		1.8%		1.3%		1.0%	
N		2		2		2		2		2		2	

Table 3 Confirmation of LLOQ in K2 EDTA rat plasma

2439-exNA		Cal-Level-1			Cal-Level-2		
		0.0 ng/mL			1.0 ng/mL		
Run ID	Sample No.	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of LLOQ
AN400.0-24_01_AB07	12, 13	0.0000	BQL	0.0%	0.0272	1.1	110.0%
AN400.0-24_01_AB07	48, 49	0.0000	BQL	0.0%	0.0259	1.0	100.0%
Mean		0.00000	n.a.	0.0%	0.02655	1.05	105.0%
SD (+/-)		0.00000	n.a.		0.00092	0.07	
CV (%)		n.a.	n.a.		3.5%	6.7%	
N		2	2		2	2	

Table 4 Selectivity in K2 EDTA rat plasma

2439-exNA					Blank		QC-LLOQ	
					0.0 ng/mL		1.0 ng/mL	
Run ID	Date	Sex	Lot #	Sample No.	Peak Area [mV*min]	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL
AN400.0-24_01_AB07	31-Jul-2024	male	RAT525364	21, 24	0.0000	0.0%	0.0233	1.0
	31-Jul-2024	female	RAT525367	22, 25	0.0000	0.0%	0.0274	1.1
	31-Jul-2024	female	RAT525368	23, 26	0.0000	0.0%	0.0279	1.1

Table 5 Carry-over in K2 EDTA rat plasma

2439-exNA		Cal-Level-2			Cal-Level-7			Carry Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	Sample No.	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ
AN400.0-24_01_AB07	13, 18, 19	0.0272	1.1	110.0%	36.5243	992.8	99.3%	0.0000	BQL	0.0%
AN400.0-24_01_AB07	49, 54, 55	0.0259	1.0	100.0%	37.0626	1,007.4	100.7%	0.0000	BQL	0.0%

Table 6 Accuracy and Precision in K2 EDTA rat plasma

2439-exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	Sample No.	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal
AN400.0-24_01_AB07	24, 27, 30, 33	1.0	100.0%	2.5	83.3%	473.0	105.1%	933.7	103.7%
	25, 28, 31, 34	1.1	110.0%	2.6	86.7%	464.9	103.3%	931.9	103.5%
	26, 29, 32, 35	1.1	110.0%	2.7	90.0%	443.4	98.5%	930.1	103.3%
Mean		1.07	106.7%	2.60	86.7%	460.43	102.3%	931.90	103.5%
SD (+/-)		0.06		0.10		15.30		1.80	
CV (%)		5.4%		3.8%		3.3%		0.2%	
N		3		3		3		3	

Table 7 Freeze and thaw and short term stability in K2 EDTA rat plasma

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_01_AB07	31-Jul-2024	41, 44	3 Cycles (- 80°C) 4 hours on wet ice	2.8	93.3%	954.0	106.0%
	31-Jul-2024	42, 45		3.0	100.0%	951.1	105.7%
	31-Jul-2024	43, 46		2.8	93.3%	970.3	107.8%
Mean				2.87	95.6%	958.47	106.5%
SD (+/-)				0.12		10.35	
CV (%)				4.0%		1.1%	
N				3		3	

Table 8 Dilution Integrity in K2 EDTA rat plasma

2439-exNA		Dilution 1:1,000 500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_01_AB07	37	533,185.3	106.6%
	38	520,220.4	104.0%
	39	526,154.8	105.2%
Mean		526,520.17	105.3%
SD (+/-)		6,490.17	
CV (%)		1.2%	
N		3	

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- For preparation of calibration and QC-standards first mix 10 µL of blank plasma with a mixture of [redacted]

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6 RECORDS

The following records in addition to the above-mentioned Forms, Master or Spreadsheet Applications are required in this process:

- N/A

7 CHANGE HISTORY

Edition, Section(s)	Change	Rationale
Ed 01	New document	Assay description after the method development